[INCH-POUND]
A-A-52502B
May 11, 2010
SUPERSEDING
A-A-52502A
January 16, 1998

COMMERCIAL ITEM DESCRIPTION

VALVE ASSEMBLY, POPPET, HULL DRAIN-VEHICULAR, 3 INCH

The General Services Administration has authorized the use of this commercial item description (CID) for all Federal agencies.

1. SCOPE. This CID covers requirements for a 3 inch (in.) hull drain poppet valve assembly, which is used to drain water out of vehicles. The valve assembly consists of a valve, seat, spring, and knob.

2. SALIENT CHARACTERISTICS

2.1 Materials. Material shall be as specified herein or of equivalent strength and characteristics. The use of recovered material made in compliance with regulatory requirements is acceptable provided that all requirements of this CID are met (see 3.1).

2.1.1 Seat and valve. The seat and the valve shall be made of aluminum, Unified Numbering System (UNS) A03550, A03560, or A96061 (tempered T6). Both machined and unmachined surfaces shall be anodized.

2.1.2 Knob. The knob shall be made of steel, UNS G10100 to UNS G10250, and zinc plated in accordance with ASTM B633.

2.1.3 Spring. The spring shall be made of corrosion resistant steel wire in accordance with ASTM A313/A313M.

2.2 Design and construction. The valve assembly shall be designed and constructed in accordance with figures 1 thru 4.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to DAMI_STANDARDIZATION@conus.army.mil or U.S. Army RDECOM, Tank Automotive Research, Development and Engineering Center, ATTN: RDTA-EN/STND/TRANS MS #268, 6501 E. 11 Mile Road, Warren, MI 48397-5000. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.daps.dla.mil/online/.

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2.3 Identification markings. Identification markings shall be permanent and legible and shall include, as a minimum, the manufacturer's identification code (CAGE), the contract number, and the part or identifying number (PIN) (see 6.1).

3. REGULATORY REQUIREMENTS

3.1 Recovered material. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4. PRODUCT CONFORMANCE PROVISIONS

4.1 Responsibility for inspection. The contractor is responsible for the performance of all inspections (examinations and tests).

4.2 Product conformance. The products shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The Government reserves the right to require proof of such conformance.

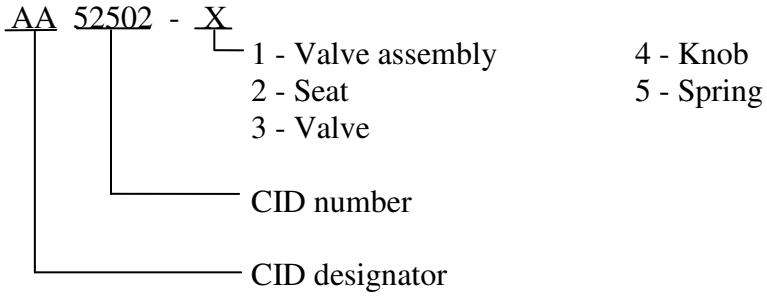
5. PACKAGING

Preservation, packaging, packing, labeling, and marking for the desired level shall be as specified in the contract (see 6.3).

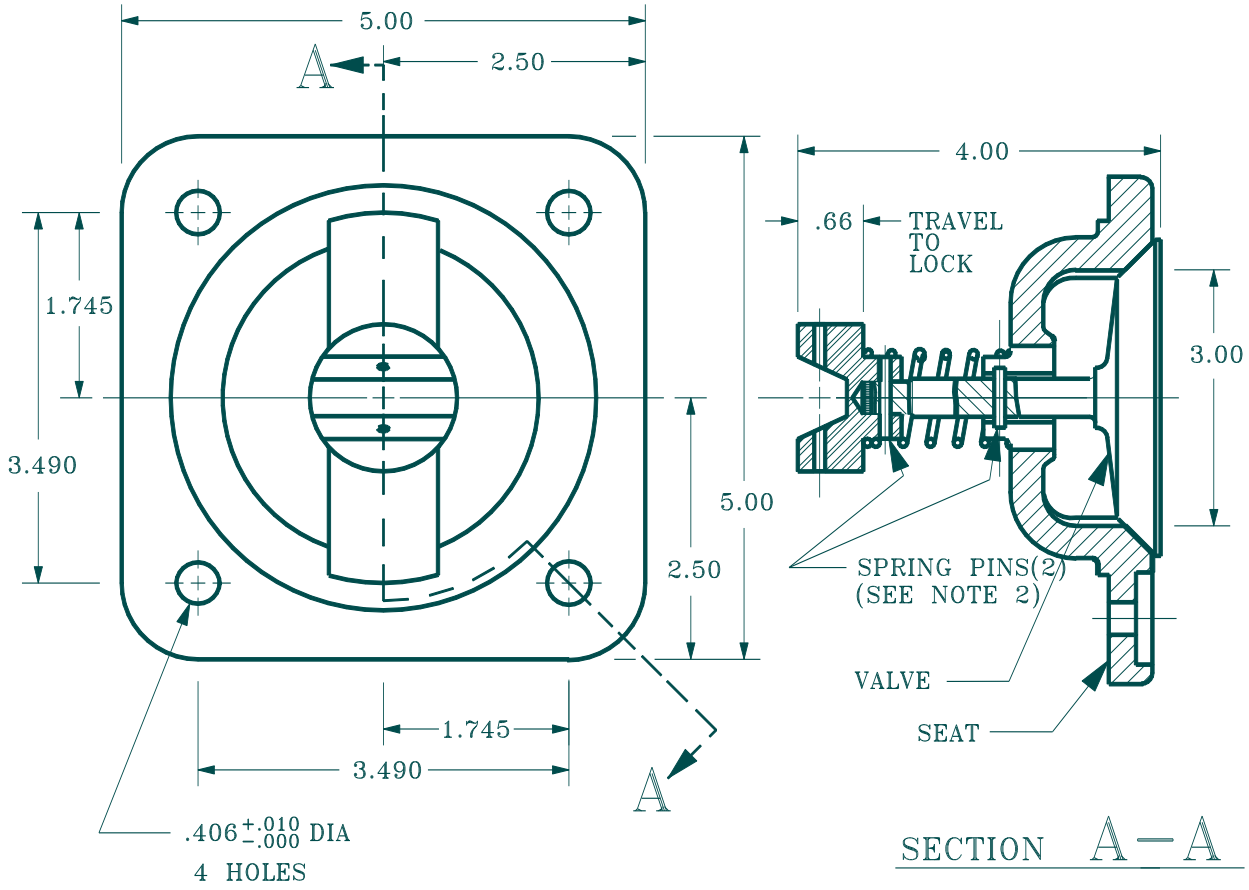
6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Part or identifying number (PIN). The following part identification numbering system is for Government purposes and does not constitute a requirement for the contractor. The PINs to be used for the valve assembly acquired to this CID are created as follows:



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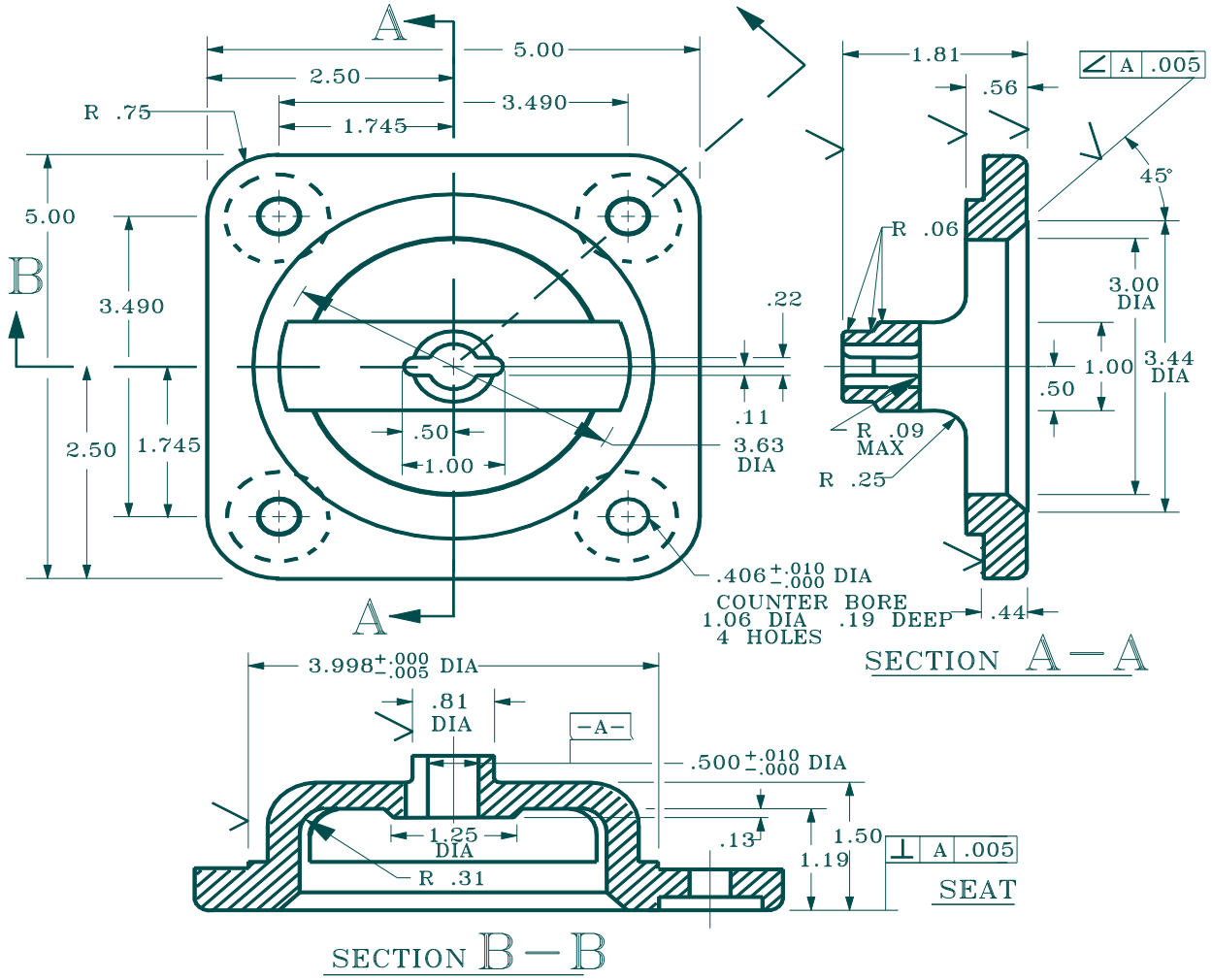
PIN	Former MS part number
AA52502-1	MS51345-2

NOTES:

1. Dimensions shown are for engineering reference only.
2. Corrosion resistant, steel spring pins with 0.125 in. nominal diameter and a length of 0.750 +0.015 in. (see 6.5).

FIGURE 1. Valve assembly.

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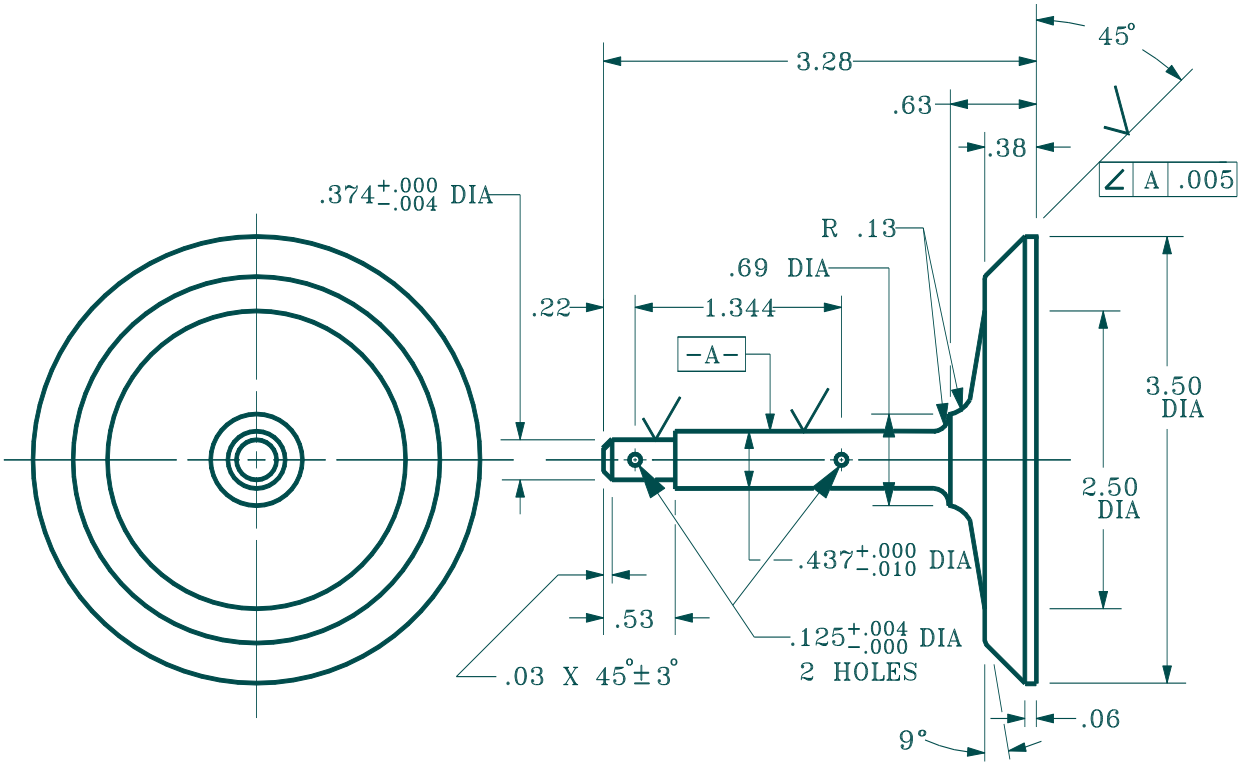
PIN	Former MS part number
AA52502-2	MS51344-2

NOTES:

1. Dimensions are in inches. Unless otherwise specified, tolerances are: Machined surfaces 2 decimals ± 0.015, 3 decimals ± 0.005, angles ± 0° 30'. Unmachined surfaces 2 decimals ± 0.031, angles ± 1°.
2. Unless otherwise specified, all machined surfaces shall be 125 microinches where roughness symbol is shown. All burrs and sharp edges shall be removed.

FIGURE 2. Seat.

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VALVE

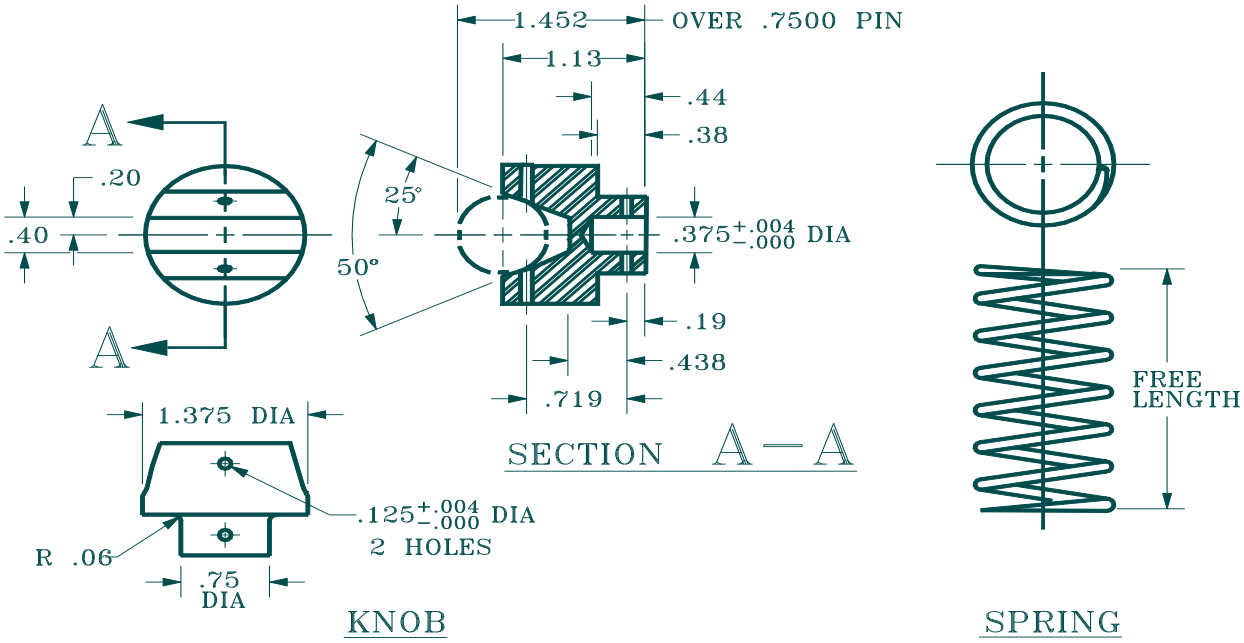
PIN	Former MS part number
AA52502-3	MS51343-2

NOTES:

1. Dimensions are in inches. Unless otherwise specified, tolerances are: Machined surfaces 2 decimals ± 0.015, 3 decimals ± 0.005, angles ± 0° 30'. Unmachined surfaces 2 decimals ± 0.031, angles ± 1°.
2. Unless otherwise specified, all machined surfaces shall be 125 microinches root mean square (rms) where surface roughness symbol is shown. All burrs and sharp edges shall be removed.

FIGURE 3. Valve.

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PIN	Former MS part number
AA52502-4	MS51349-1

PIN	Former MS part number
AA52502-5	MS51350-1

SPRING SPECIFICATION

DIAMETER OF WIRE	0.095
DIAMETER OF COIL (INSIDE)	0.91 ± 0.03
FREE LENGTH	2.360 ± 0.020
ACTIVE COILS	6.25
TOTAL COILS	7.50
DIRECTION OF HELIX	OPTIONAL
LOAD AT COMPRESSED LENGTH OF 1.03	16 ± 1.5 POUNDS (LB)
SPRING RATE	12 LB/IN
SOLID LENGTH	0.81 MAX
TYPE OF ENDS	CLOSED ENDS GROUND
SQUARENESS OF ENDS IN FREE POSITION WITHIN 3°	

NOTES:

1. Dimensions are in inches. Unless otherwise specified, tolerances are:
   a. Knob tolerances are: on 2 decimals ± 0.015, on 3 decimals ± 0.005, angles ± 0°30'.
   b. Spring tolerances are: on 2 decimals ± 0.031, on 3 decimals ± 0.020.
2. All knob machined surfaces shall be 125 microinches. All burrs and sharp edges shall be removed.

FIGURE 4. Knob and spring.

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6.2 Addresses for obtaining copies of referenced documents.

6.2.1 Government publications. The Federal Acquisition Regulation (FAR) may be obtained at the following website: http://www.acquisition.gov/far.

6.2.2. Non-Government publications. Copies of SAE-HS-1086 "Metals and Alloys in the Unified Numbering System" are available from the Society of Automotive Engineers, Inc. (SAE), 400 Commonwealth Drive, Warrendale, PA 15096-0001 or www.sae.org; Copies of ASTM A313/A313M "Standard Specification for Steel Wire, Music Spring Quality" (DoD Adopted), and ASTM B633 "Standard Specification for Electrodeposited Coatings of Zinc on Iron and Steel" (DoD Adopted) are available from the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959 or www.sae.org.

6.3 Ordering data. Acquisition documents must specify the following:

a. Title, number, and date of this CID.
b. PIN number and quantity required.
c. Packaging requirements (see 5).
d. Issue of Department of Defense Index of Specifications and Standards (DoDISS) to be cited in the solicitation and if required the specific issue of individual documents referenced.

6.4 Cross-reference data. Valve assemblies and components are interchangeable/substitutable with valve assemblies and components conforming to MS51343B(AT), MS51344C(AT), MS51345B(AT), MS51349A(AT), and MS51350A(AT).

6.5 Replacement spring pins. For replacement spring pins, use NASM171401 thru NASM171900, part no. MS171530.

6.6 Key words.

Knob
M9 ACE
Pin
Seat
Spring
Water

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MILITARY ACTIVITY:

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Army - AT

Preparing Activity:
Army - AT

Review Activity:
DLA - CC

(Project 2590-2010-003)

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